

1. Analysis and Insights

Data Filter: focused on the most recent complete year (**2023**).

- **Basket of Goods:** Selected three representative commodities to act as a "Basic Food Plate":
 - **Carbohydrate:** Rice (Well Milled)
 - **Protein:** Fish (Milkfish/Bangus)
 - **Vegetable:** Eggplants
- **Algorithm:** Applied **K-Means Clustering** (with $k=3$) to group the 73 markets based on the average prices of these items.

Cluster Profiles:

The algorithm identified three distinct types of markets in the Philippines:

Cluster Name	Rice Price (Avg)	Fish Price (Avg)	Veg Price (Avg)	Primary Characteristic
Cluster A (Budget-Friendly)	₱44.49	₱199.24	₱63.44	Lowest prices across all categories.
Cluster B (High-Staple Cost)	₱49.13	₱224.62	₱86.01	Most expensive Rice and Fish.
Cluster C (High-Veg Cost)	₱44.99	₱211.71	₱94.83	Significantly higher vegetable prices (+50% vs Cluster A).

2. Summary of Findings

The clustering reveals a clear geographic divide in food affordability:

1. **Mindanao is the Most Affordable (Cluster A):**

- Markets like **Agusan del Norte, Bukidnon, and Davao (Region XI)** fall into the "Budget-Friendly" cluster.
- These regions likely benefit from close proximity to agricultural production, keeping transport costs low for both crops and fish.

2. **The Visayas Struggle with Staples (Cluster B):**

- Regions such as **Central Visayas (Cebu, Bohol)** and **Eastern Visayas** are in the "High-Staple Cost" cluster.
- They pay the highest prices for Rice and Fish, likely due to higher logistics costs involved in inter-island shipping.

3. **Luzon & Western Visayas Face Vegetable Inflation (Cluster C):**

- Markets in **Bicol, Western Visayas, and Central Luzon** fall into the "High-Veg Cost" cluster.
- Despite being on major islands, these areas pay nearly **₱30 more per kg** for vegetables than the budget cluster, suggesting inefficiencies in the cold chain or distribution networks for perishable goods.

3. Recommendations

Based on these clusters, here are targeted recommendations:

• **For Cluster B (Visayas - High Staple Costs):**

- **Policy:** Implement targeted rice and fisheries subsidies or "Kadiwa" stores specifically for basic staples.
- **Infrastructure:** Invest in port infrastructure and roll-on/roll-off (RoRo) transport efficiency to lower the shipping cost of rice and fish from Mindanao/Luzon to the Visayas.

• **For Cluster C (Luzon/W. Visayas - High Veg Costs):**

- **Supply Chain:** Focus on **Cold Chain Storage** facilities. The high cost of perishables like eggplants suggests high spoilage rates during transport.
- **Agriculture:** Encourage urban gardening or local sourcing initiatives to reduce dependency on long-distance vegetable transport.

• **For Cluster A (Mindanao - Budget Friendly):**

- **Monitoring:** Use these markets as a "benchmark" for national price stability.
- **Protection:** Ensure that as these regions develop, their local agricultural advantages are protected from rapid urbanization that might displace farmlands.